

Numerical Analysis of the Ignition Transient of Solid Rocket Motors

Master's Thesis in Space and Astronautical Engineering



SAPIENZA
UNIVERSITÀ DI ROMA

Giorgia Mazzotta
ID number: 1938281

Advisor
Prof. Daniele Bianchi

Co-Advisor
Dr. Marco Grossi

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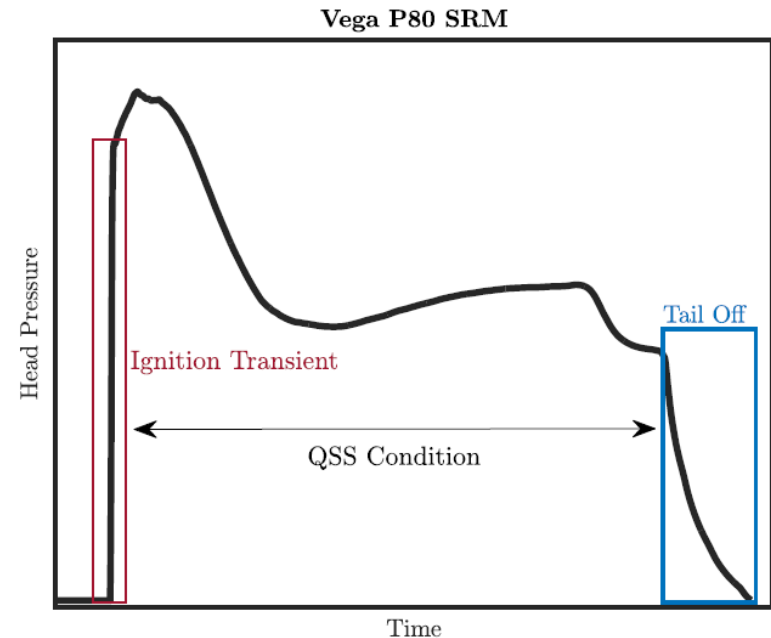
❑ Final Remarks

SRMs Ignition Transient: Phenomenology

The Ignition Transient (IT) is the time interval between the first signal of the ignition chain for the motor start-up and the attainment of the quasi-steady-state (QSS) condition of the SRM.

Ignition Transient Sequence:

- ❑ Electric Delay
- ❑ Induction Interval
- ❑ Flame Spreading
- ❑ Chamber Filling



The IT of a SRM may be a **critical** and/or **dimensioning phase** in the mission:

- Overpressure peaks
- Pressure oscillations & pressure unbalances
- Hangfires or misfires
- Dynamic loads on the vehicle...

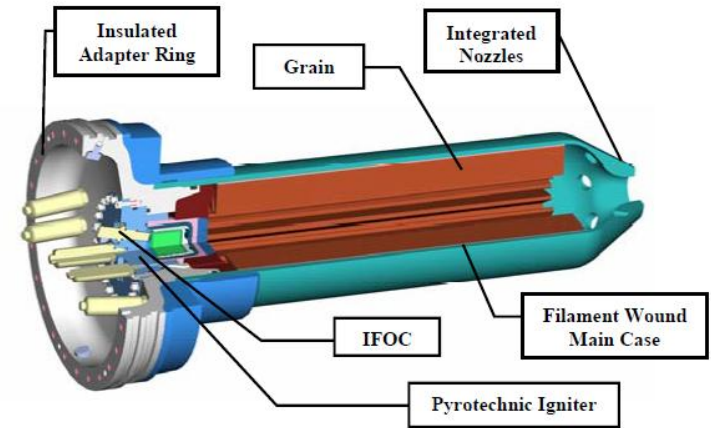
SRMs Ignition Transient: State of Art of Igniters

Role of the main igniter: generation and transfer of energy to the motor to initiate the combustion of the propellant grain and ensure the SRM ignition

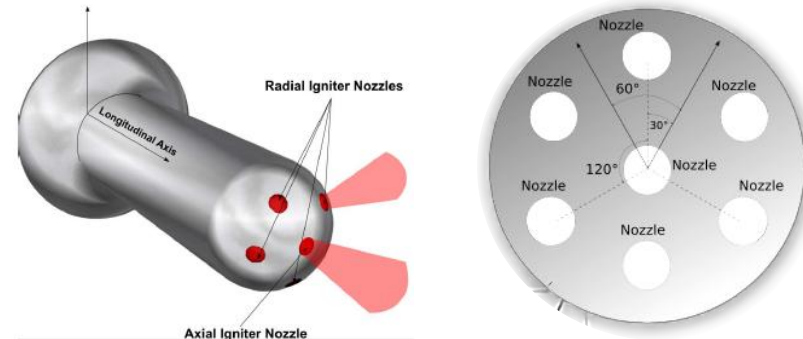
VEGA Ignition Chain

1. Initiators activate the **pyrotechnic** pellets
2. Hot flow ignites the **pyrogen** main igniter propellant grain
3. Hot combustion products flow into the SRM through **igniter nozzles**
4. **Wave propagation** phenomena due to igniter mass flow rate during **closed chamber** operation (nozzle seal)

How the **configuration** of the igniter nozzles influence the ignition transient?



Configuration of a VEGA SRM igniter



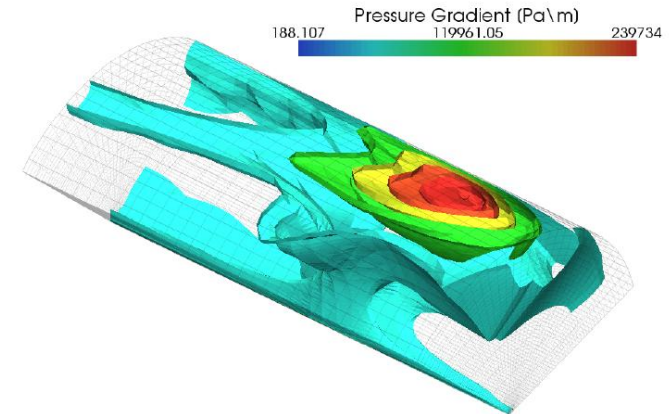
Schematic representation of the 6+1 igniter nozzles topology

Numerical Models: Computational Fluid Dynamics (CFD)

Unsteady numerical simulations based on RANS (Reynolds Averaged Navier-Stokes) equations

Key assumptions:

- **Fluid model:** multi-species, single-phase, non-reacting mixture of perfect gas;
- Explicit, **first-order** accurate scheme with CFL condition;
- **No** flame spreading for the propellant **grain ignition** and burning → negligible effects on pressure oscillations;
- **No opening** dynamics of the main **nozzle seal**.



Example of a CFD simulation: Zefiro 16
3D pressure gradient*

*Zaghi, S. et al, "3D simulations of pre-Ignition Transient of P80 SRM", 44th AIAA/ASME/SAE/ASEE Joint Propulsion Conference & Exhibit, 2008.

GOAL: Investigate the igniter jets structure and the resulting multidimensional flow field inside the motor port volume

Numerical Models: SPIT (Solid Propellant rocket Ignition Transient)

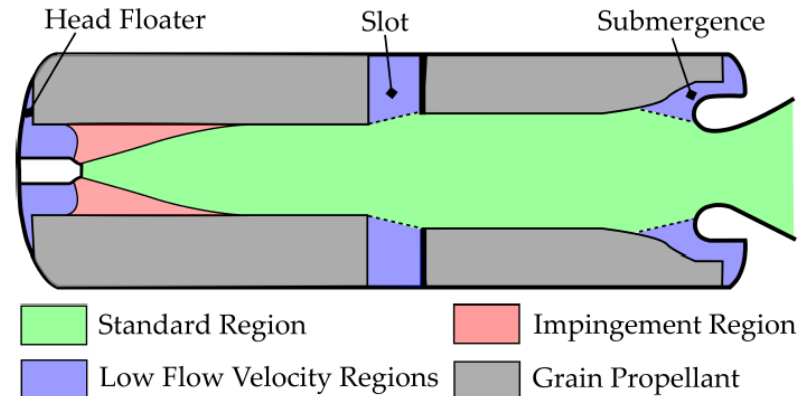
- **Quasi-1D unsteady** model for a non-reactive mixture of perfect gases with **source terms** due to grain propellant, igniter, cavity regions and friction

Need for further sub-models:

- Igniter sub-model
- Grain sub-model
- Combustion sub-model
- **Cavity sub-model**



Coupling between the main flow and cavities through mass exchange



Effective cavity pressure: $\hat{p}_{cav} = p_{cav} - \alpha_{cav}(p_{cav} - p_{cham})$

Energy balance within the cavity: $\Delta H = -(H_{cav} + H_{pr} - \hat{H}_{cav})$ $\longrightarrow$

Its sign determines the direction of the flow across the cavity/chamber interface

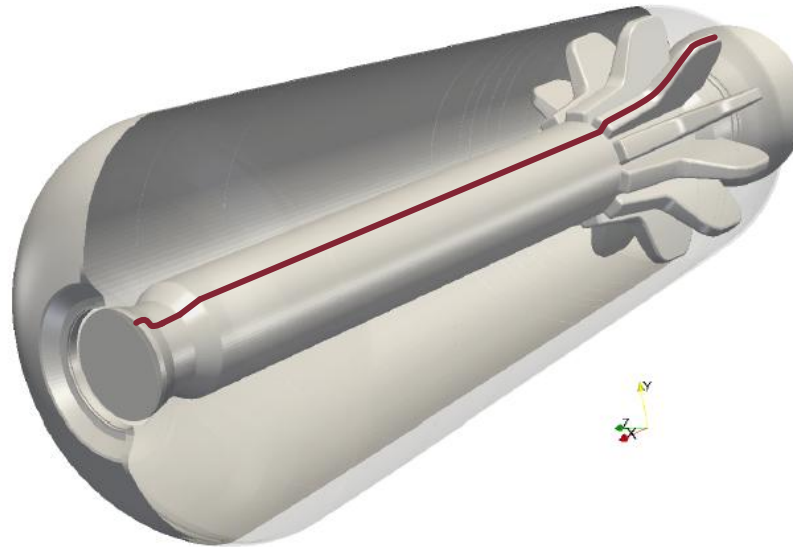
GOAL: cavity sub-model **calibration** and SPIT **validation** against CFD and experimental data

CFD Analysis: Computational Domain

- **2D axisymmetric** simulation of the P80 SRM (VEGA First-Stage)
- Multiblock structured grid developed using the open-source tool GMSH



Chamber free volume

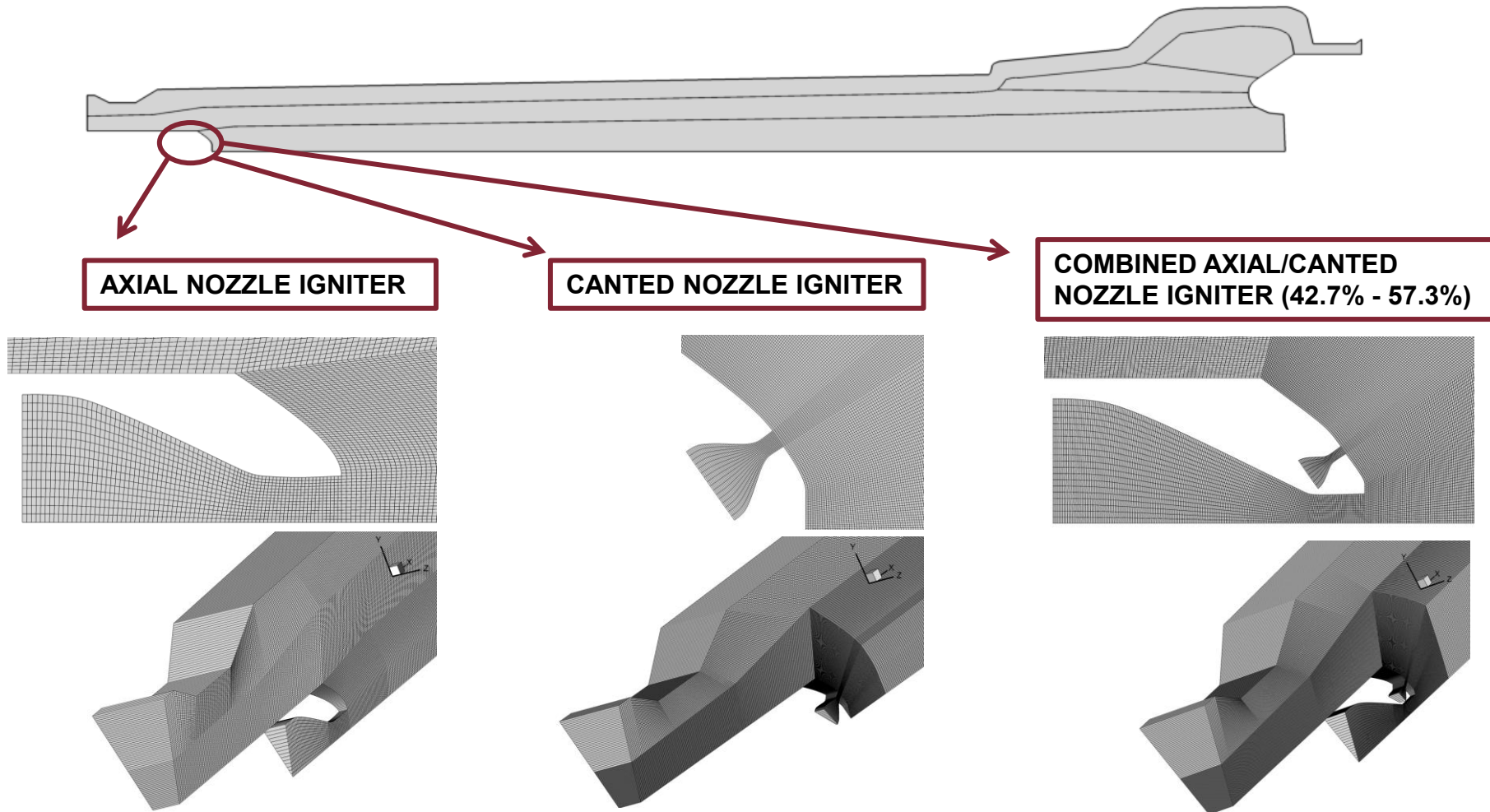


P80 propellant grain configuration*

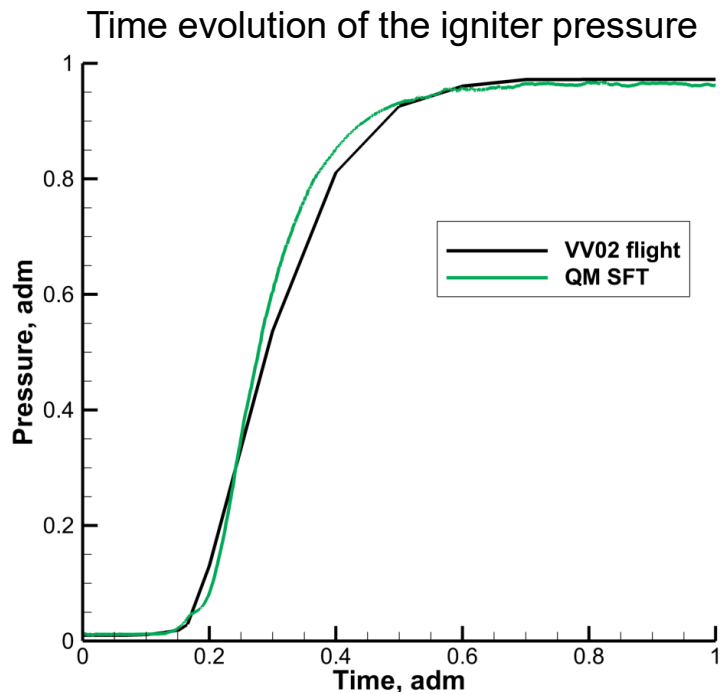
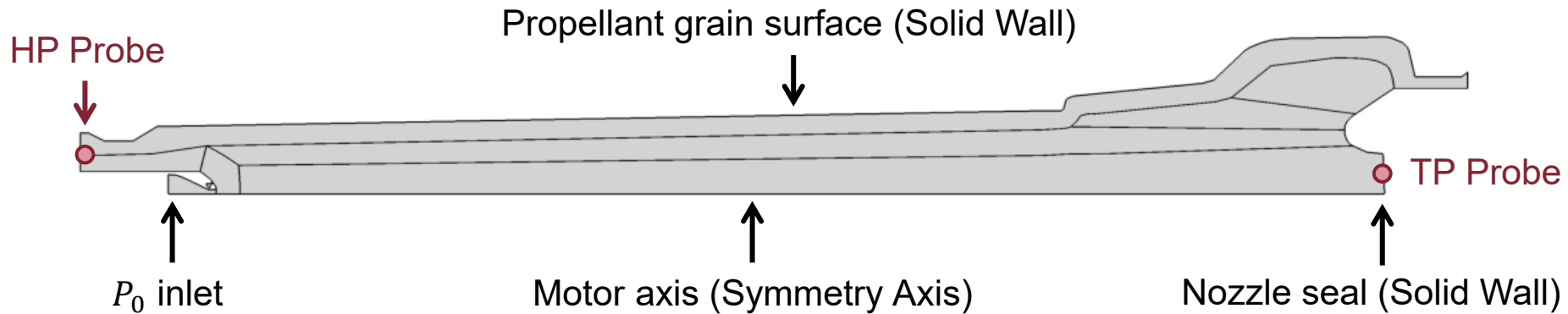
*Cavallini, E. et al, "Analysis and Performance Reconstruction of VEGA Solid Rocket Motors Qualification Flights", 50th AIAA/ASME/SAE/ASEE Joint Propulsion Conference & Exhibit, 2014.

CFD Analysis: Computational Domain

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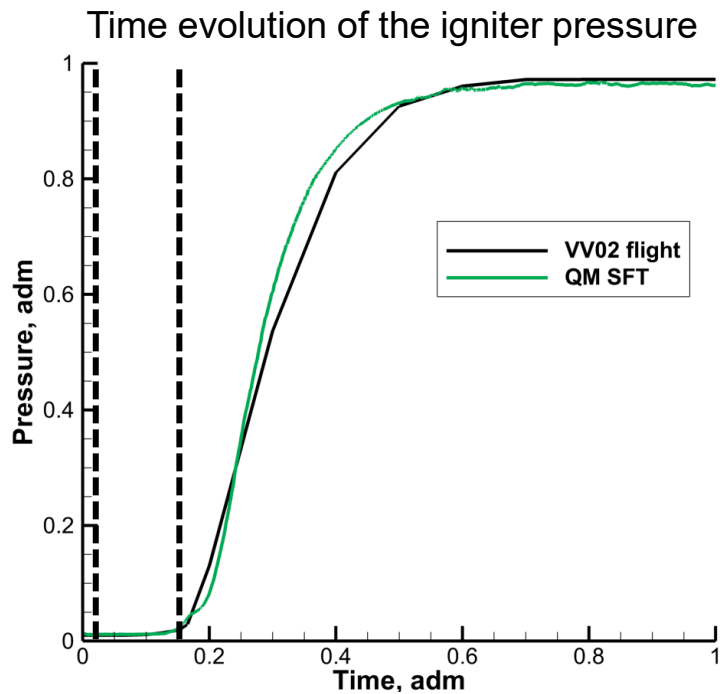
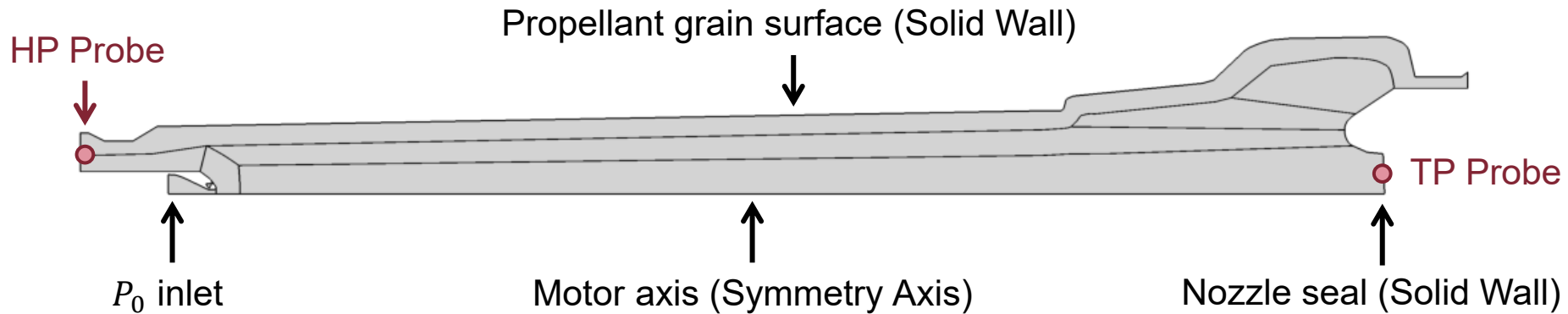
CFD Analysis: Initial and Boundary Conditions



	Igniter gas	N_2	He
P_0 [bar]	Assigned law	1.36	1.3
T_0 [K]	3366.97	299.15	299.15
γ	1.14	1.4	1.66
M [kg/mol]	27.42	28.0	4.0

CFD Test Matrix
 [3 Igniter Configurations] x [2 Pressurizing Gases (N_2 , He)]

CFD Analysis: Initial and Boundary Conditions

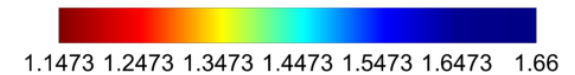


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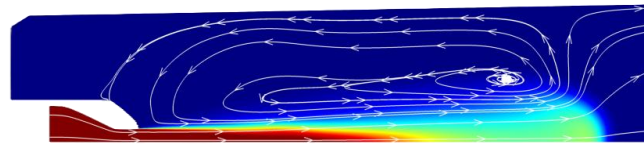
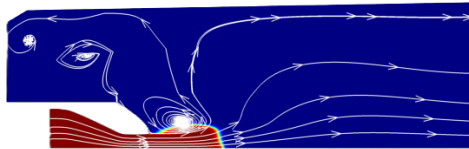
CFD Analysis: The Igniter Jets

Specific Heats Ratio



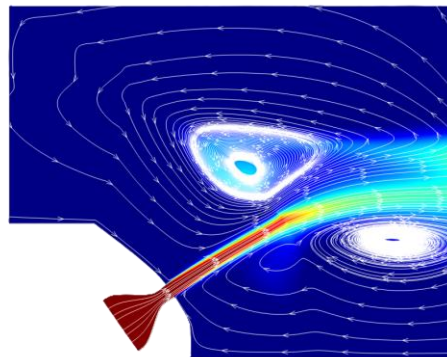
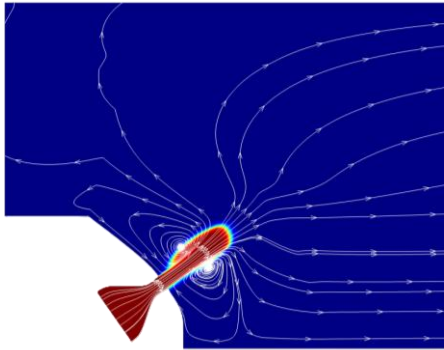
$t = 0.02$

$t = 0.15$



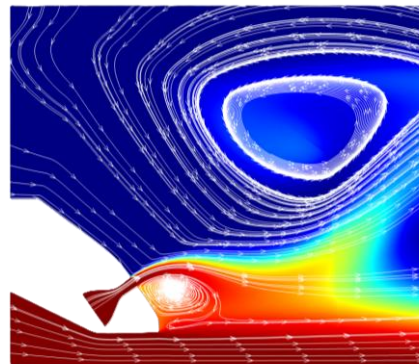
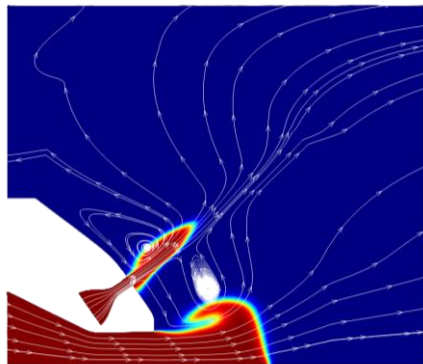
Axial Igniter

- Mixing region ahead of the jet
- Downstream progression of the recirculating zone



Canted Igniter

- Dual recirculating zones
- Bending of the jet towards the axial direction



Combined Igniter

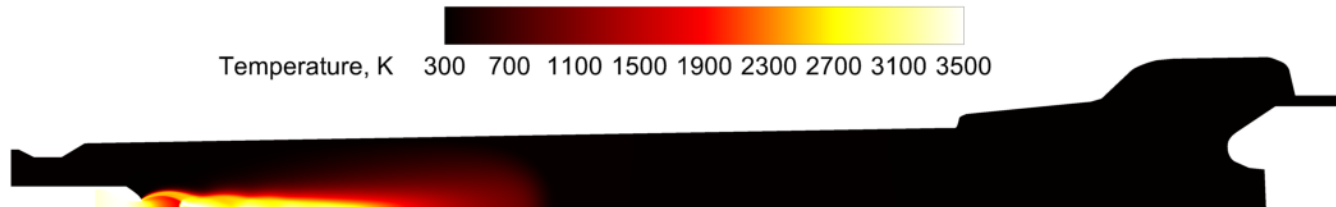
- Dual recirculating zones
- Two independent jets that rapidly merge

CFD Analysis: Temperature Field

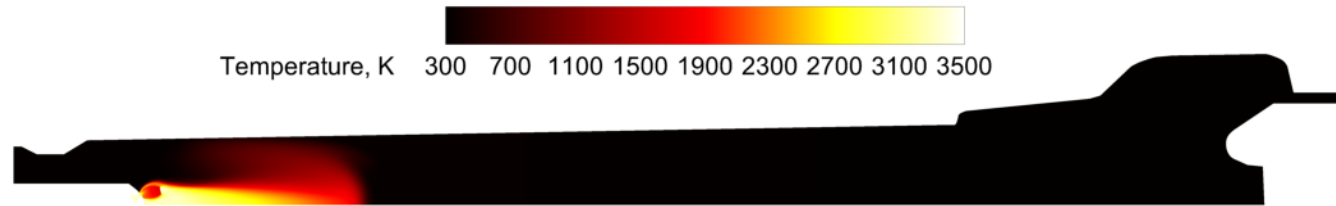
- Simulation campaign with helium as pressurizing gas

$t = 0.25$

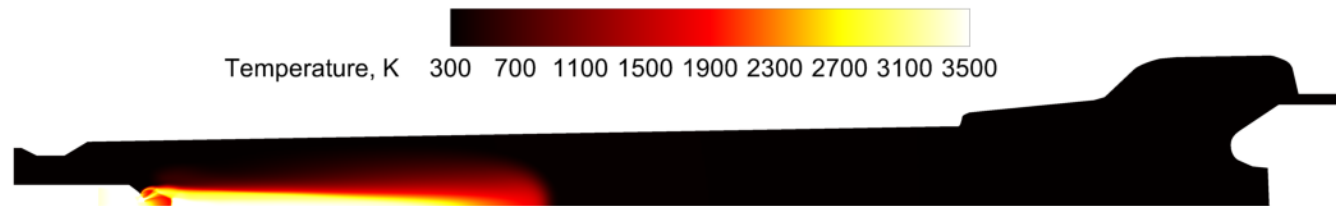
AXIAL-NOZZLE IGNITER



CANTED-NOZZLE IGNITER



COMBINED-NOZZLE IGNITER



CFD Analysis: Temperature Field

- Simulation campaign with helium as pressurizing gas

$t = 0.50$

AXIAL-NOZZLE IGNITER



CANTED-NOZZLE IGNITER



COMBINED-NOZZLE IGNITER



CFD Analysis: Temperature Field

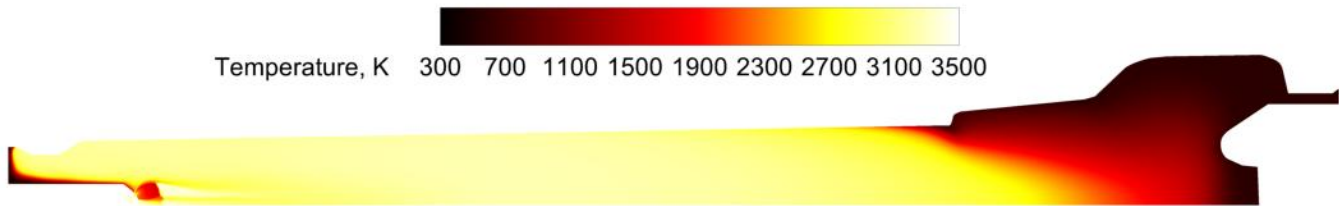
- Simulation campaign with helium as pressurizing gas

$t = 0.75$

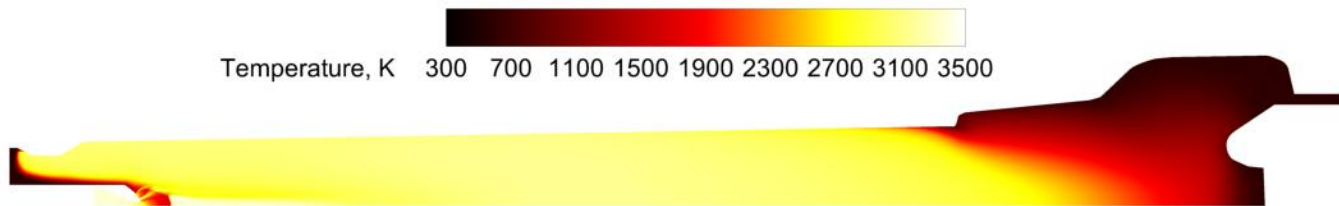
AXIAL-NOZZLE IGNITER



CANTED-NOZZLE IGNITER



COMBINED-NOZZLE IGNITER

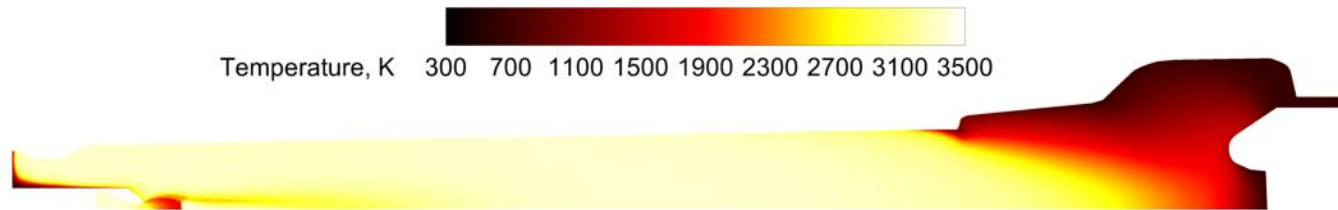


CFD Analysis: Temperature Field

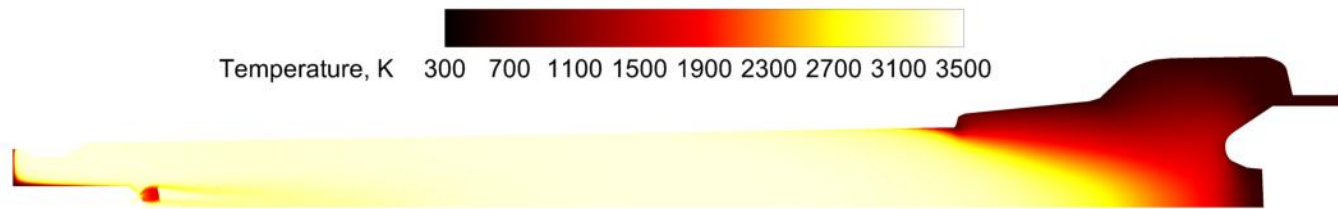
- Simulation campaign with helium as pressurizing gas

$t = 1.0$

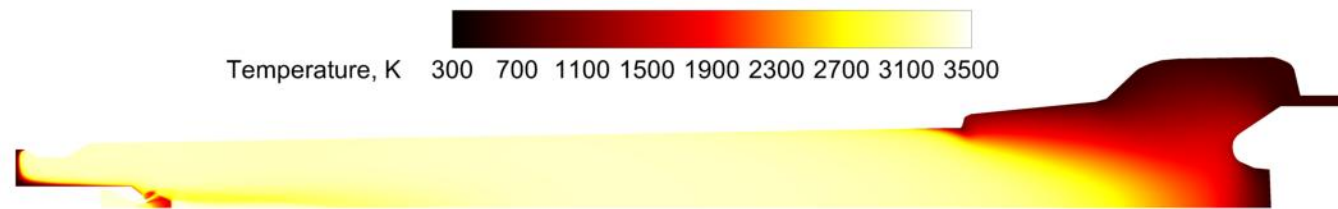
AXIAL-NOZZLE IGNITER



CANTED-NOZZLE IGNITER



COMBINED-NOZZLE IGNITER

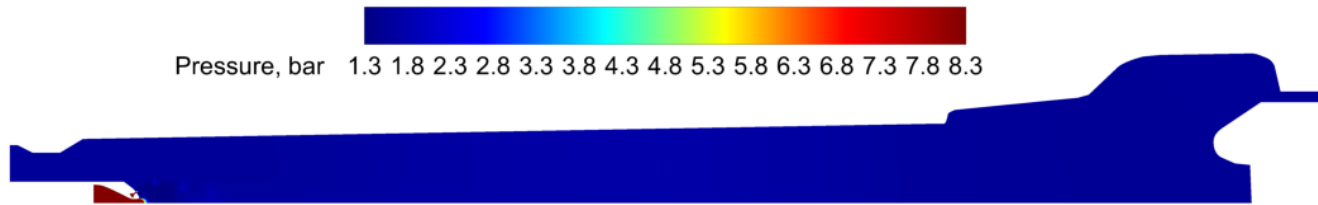


The Effect of Pressurizing Gas: Pressure Field

- Combined-igniter configuration

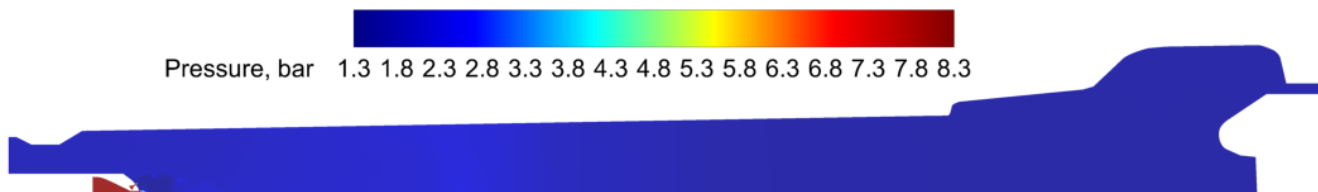
$t = 0.20$

HELIUM (He)



- High compressibility
- Low acoustic impedance
- **Acoustic properties comparable** to the igniter gas

NITROGEN (N_2)



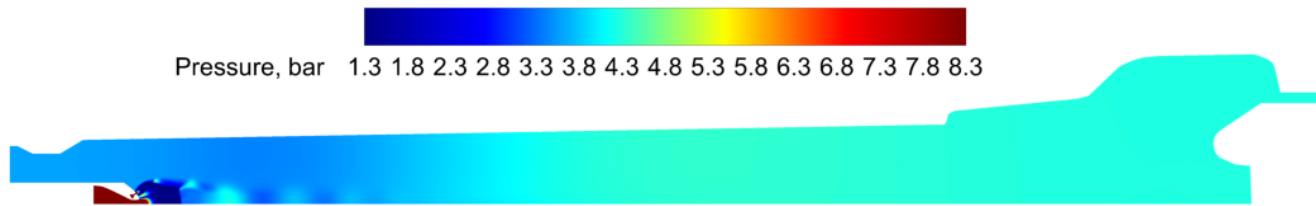
- Low compressibility
- High acoustic impedance
- **Acoustic mismatch** with the igniter gas

The Effect of Pressurizing Gas: Pressure Field

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$t = 0.50$

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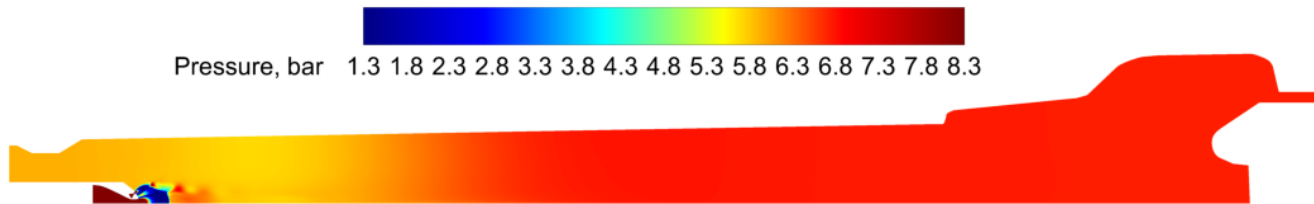
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The Effect of Pressurizing Gas: Pressure Field

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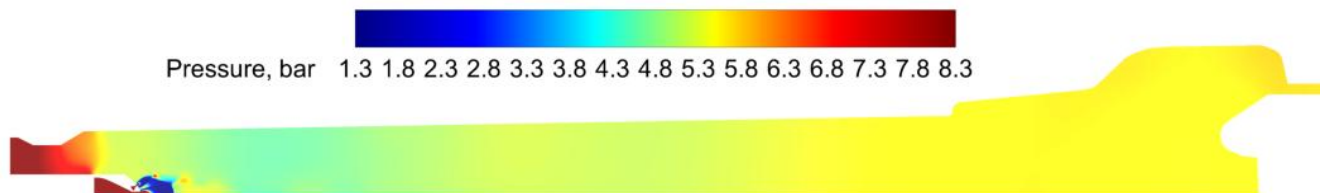
$t = 0.75$

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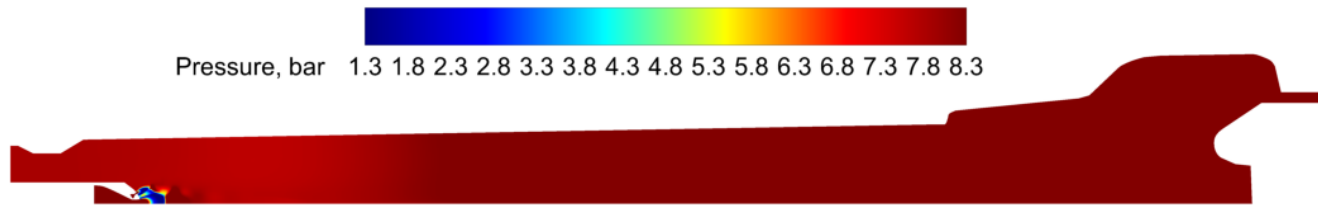
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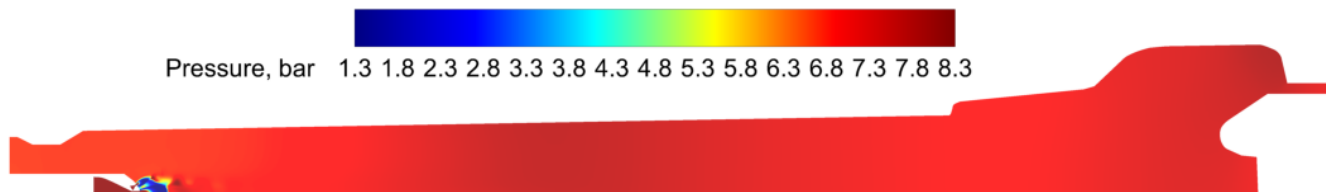
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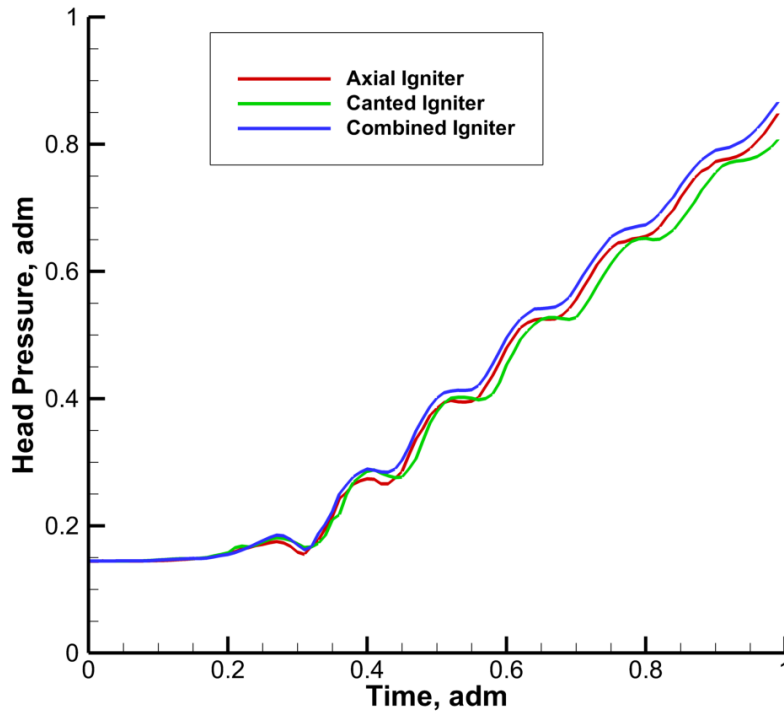
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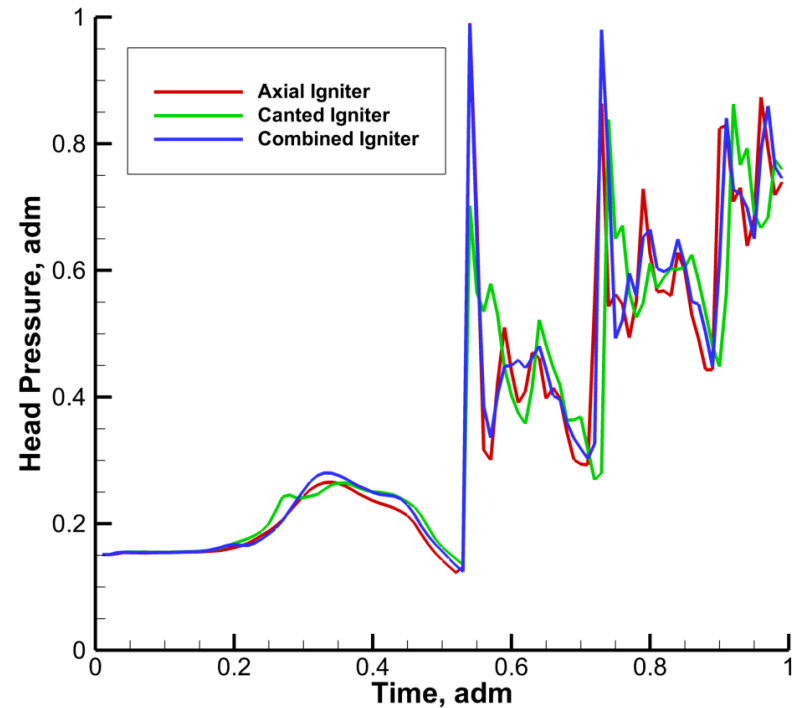
CFD Analysis Results: Head Pressure (1)

HELIUM



- Low pressure oscillations (PO)
- PO slightly affected by the igniter configuration
- Very similar pressure rise-up

NITROGEN

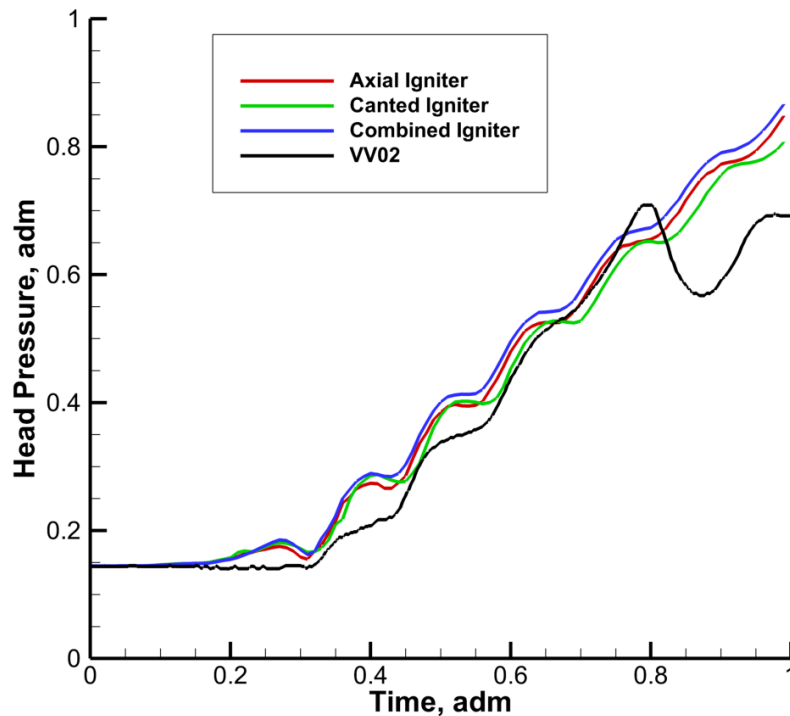


- High pressure oscillations with strong fluctuations
- PO affected by the igniter configuration
- Comparable pressure rise-up



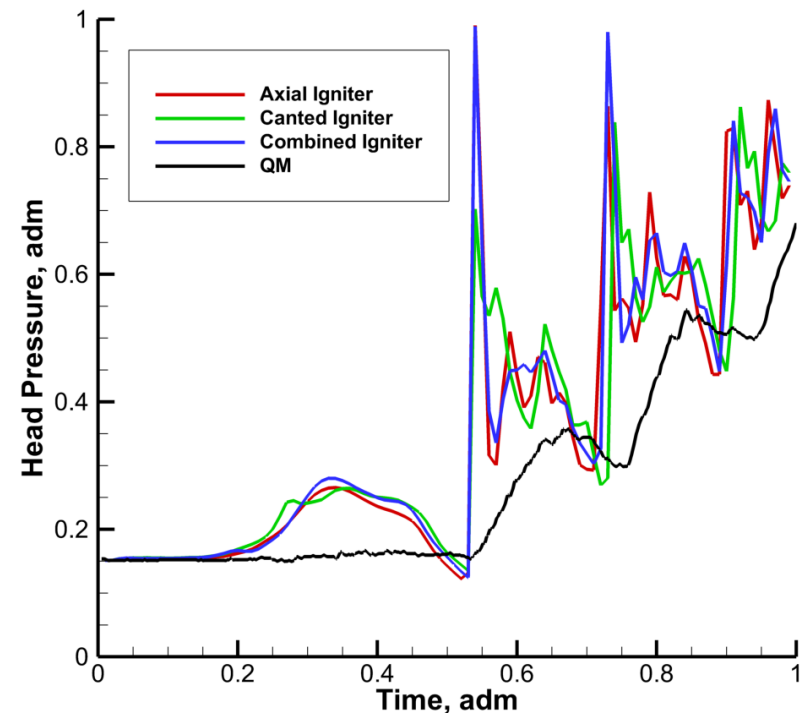
CFD Analysis Results: Head Pressure (2)

HELIUM



- Low pressure oscillations (PO)
- PO slightly affected by the igniter configuration
- Very similar pressure rise-up
- CFD simulations capture P80 SRM acoustic behaviour experienced in **flight**

NITROGEN

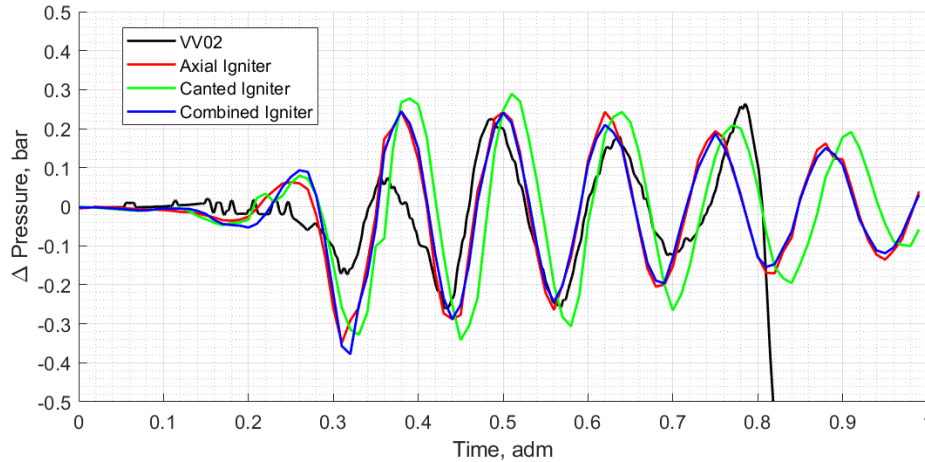


- High pressure oscillations with strong fluctuations
- PO affected by the igniter configuration
- Comparable pressure rise-up
- CFD simulations show higher level of PO and a noisier signal w.r.t. **firing test**

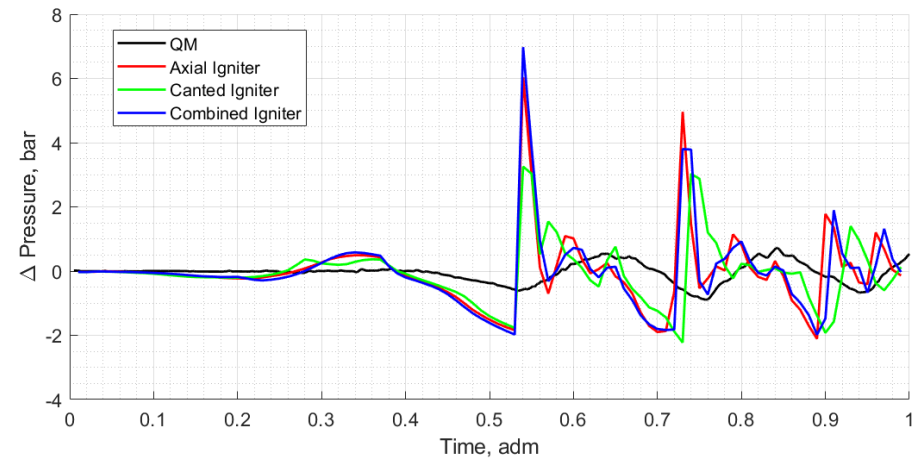


CFD Analysis Results: Head Pressure (3)

HELIUM



NITROGEN

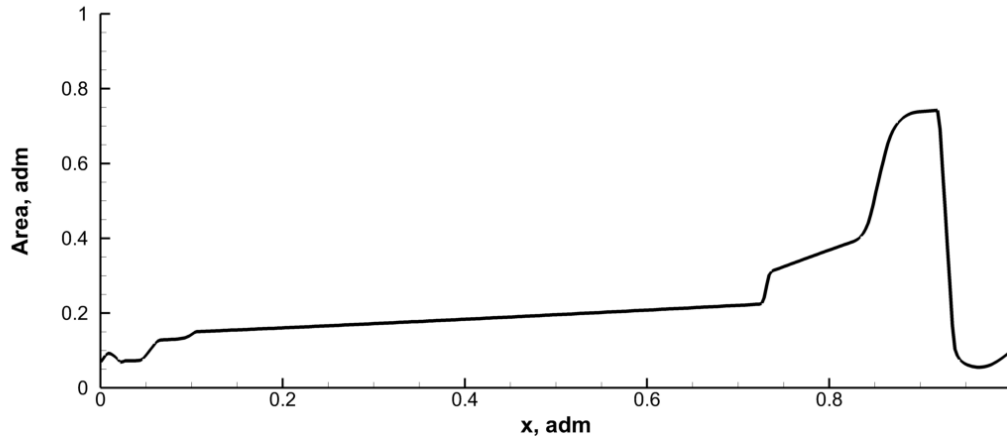


- Low pressure oscillations (PO)
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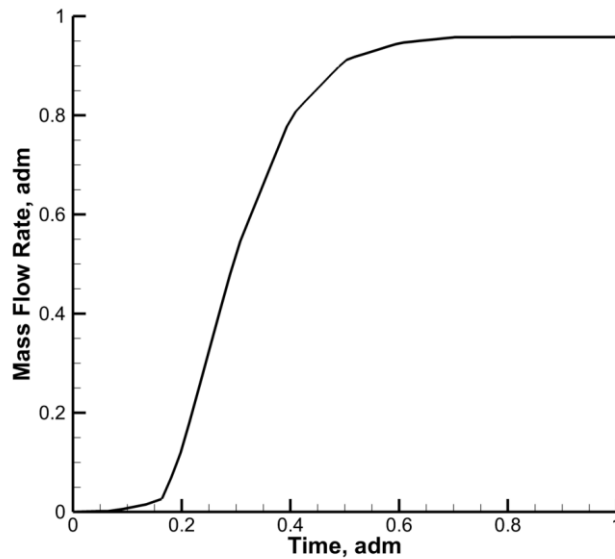
- High pressure oscillations with strong fluctuations
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Quasi-1D Analysis: Simulations Setup

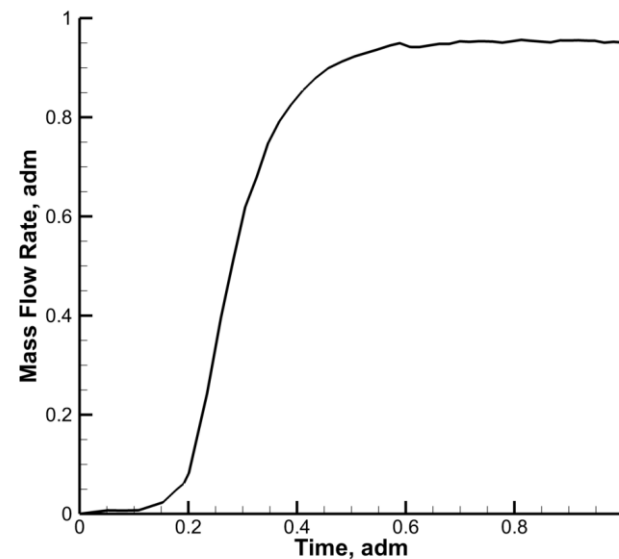


- **Initial conditions** identical to the CFD simulations
- Time-dependent **boundary conditions**: seal breakage opening, unsteady outflow conditions
- **Igniter mass flow rate** profile extracted from the CFD simulations

Helium

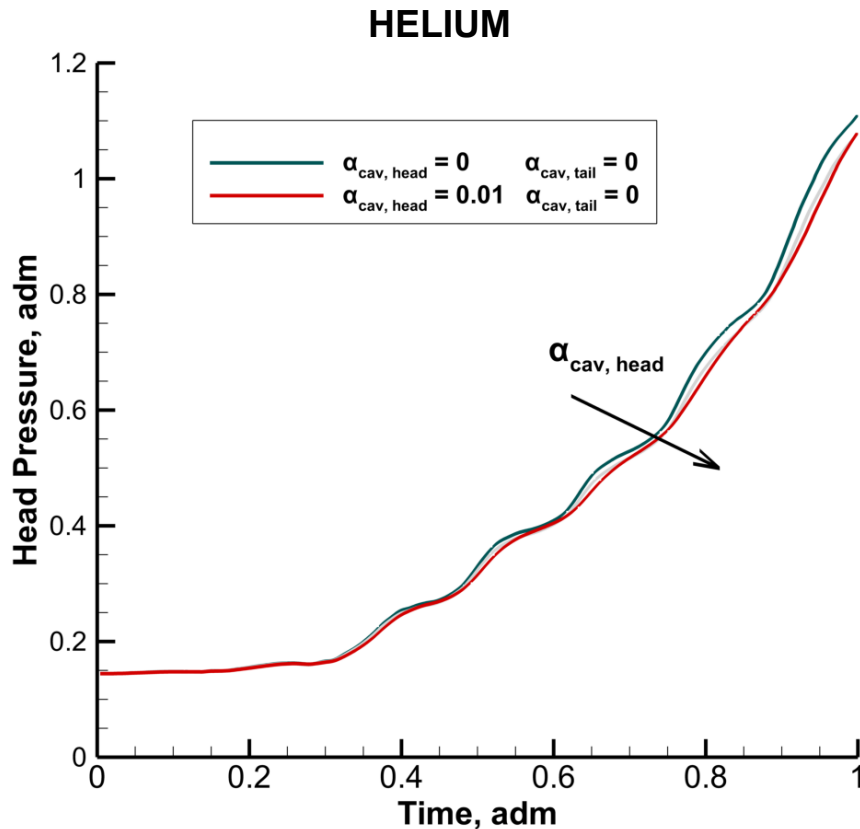


Nitrogen

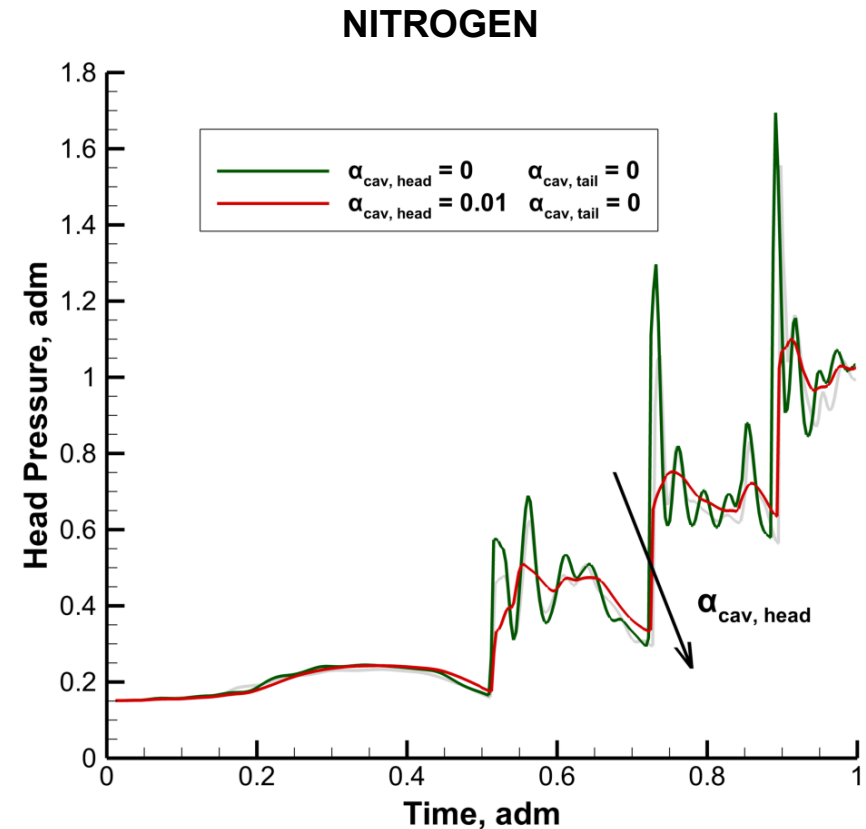


Quasi-1D Analysis: Role of the α_{cav} Parameter

- The calibration of **two** distinct **parameters** is necessary to separately model the head and tail cavities $\longrightarrow \alpha_{cav, head}$ and $\alpha_{cav, tail}$



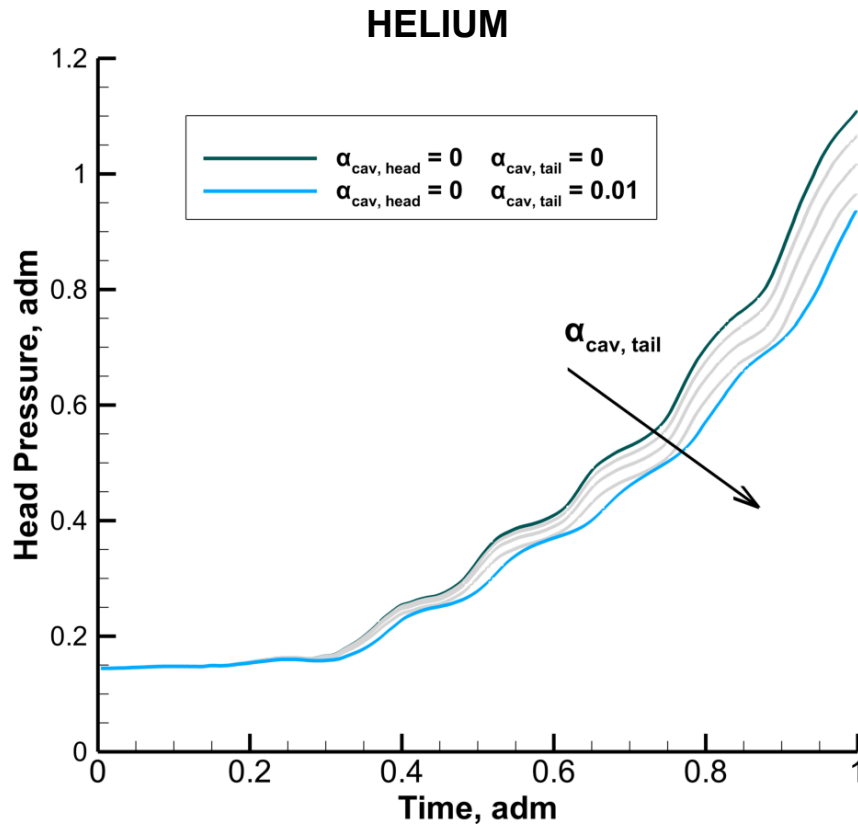
Low sensitivity to $\alpha_{cav, head}$



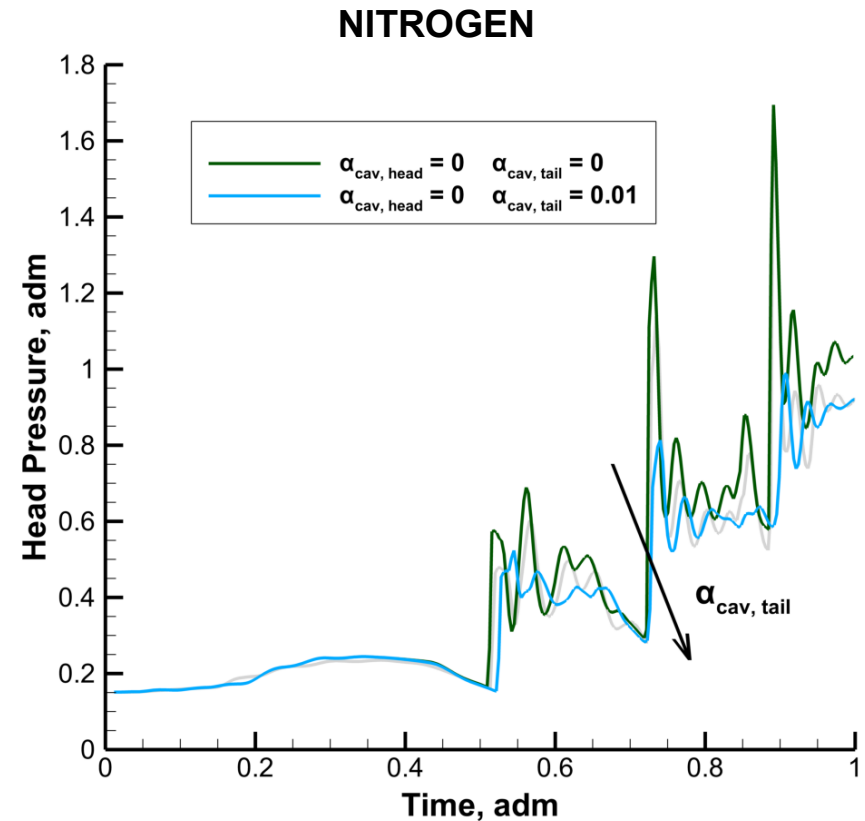
High sensitivity to $\alpha_{cav, head}$

Quasi-1D Analysis: Role of the α_{cav} Parameter

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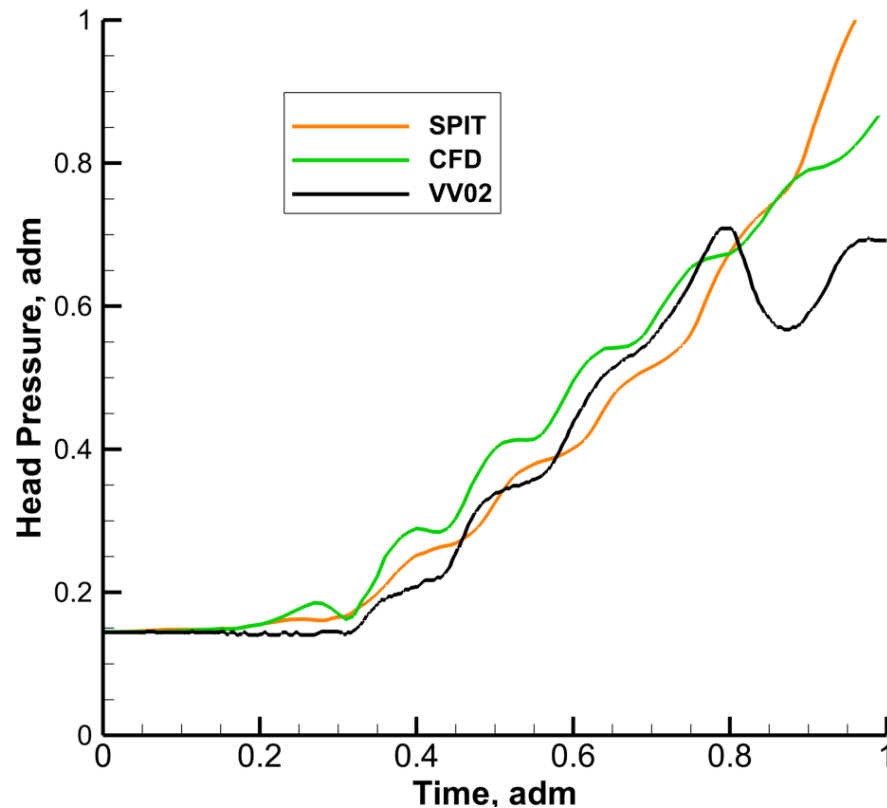
High sensitivity to $\alpha_{cav, tail}$



Medium sensitivity to $\alpha_{cav, tail}$

Quasi-1D Analysis: Helium Results

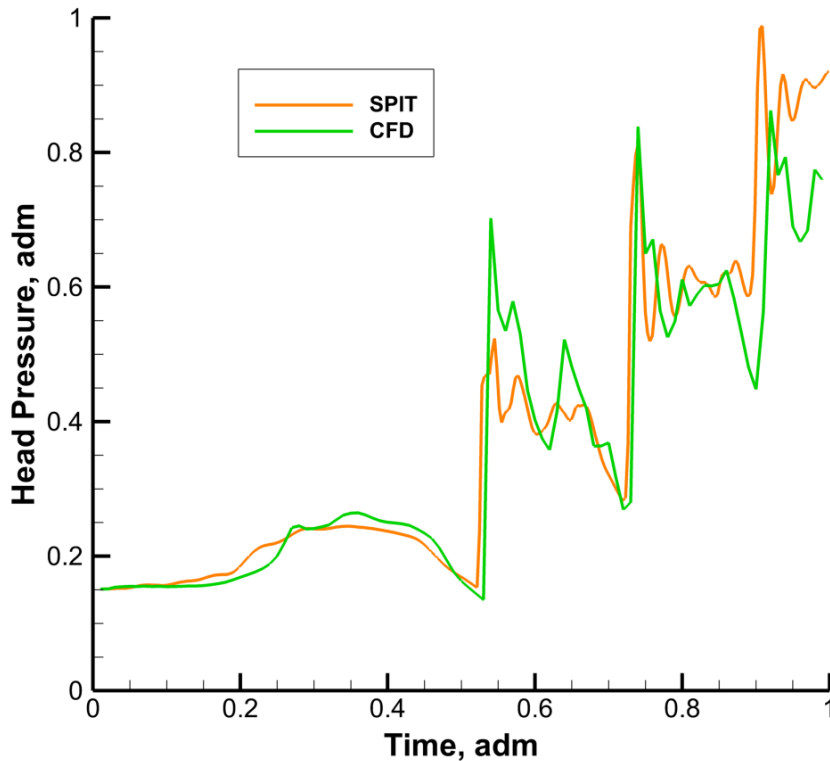
- **Combined-nozzle** igniter configuration is the CFD reference
- **Validation:** the head-end cavity model has a negligible impact for the calibration based on the experimental data



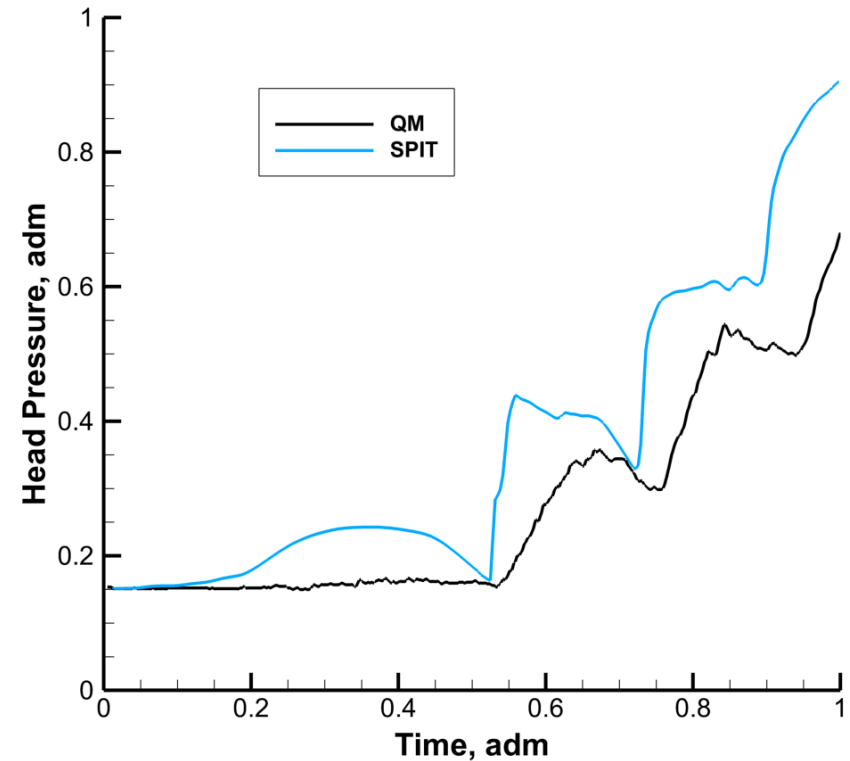
$$\alpha_{cav, head} = 0 \text{ and } \alpha_{cav, tail} = 0.001$$

Quasi-1D Analysis: Nitrogen Results

- **Canted-nozzle** igniter configuration is the CFD reference
- **Validation:** the head-end cavity model has a significant impact for the calibration based on the experimental data



$$\alpha_{cav, head} = 0 \text{ and } \alpha_{cav, tail} = 0.01$$



$$\alpha_{cav, head} = 0.01 \text{ and } \alpha_{cav, tail} = 0.01$$

Conclusions

- ✓ **Demonstrated the effectiveness of the CFD approach**
 - Accurate prediction of multidimensional igniter jet dynamics and pressurizing gas effects during the ignition transient
- ✓ **Assessed the influence of igniter nozzle configurations**
 - Severe acoustic environments may amplify differences in pressure oscillation behaviour (e.g., nitrogen vs helium)
- ✓ **Highlighted the reliability of the Quasi-1D SPIT model**
 - Properly calibrated reduced-order models can provide accurate predictive capabilities when supported by dedicated sub-models
- **Future Work**
 - 3D CFD simulations to further investigate igniter configurations
 - Integration of the head-end cavity into the CFD computational domain

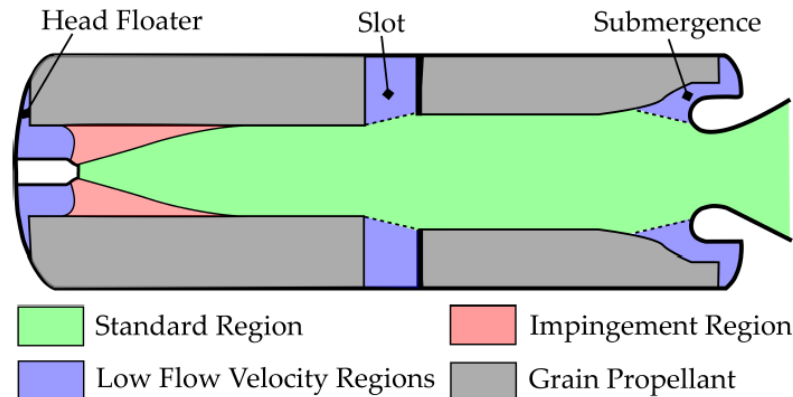
SPIT Full Model: Governing Equations

$$\left\{ \begin{array}{l} \frac{\partial(\rho_k A_p)}{\partial t} + \frac{\partial(\rho_k u A_p)}{\partial x} = \delta_{k,pr} \frac{\dot{m}_{pr} A_p}{V} + \delta_{k,ig} \frac{\dot{m}_{ig} A_p}{V} + \delta_{k,cav} \frac{\dot{m}_{k,cav} A_p}{V} \\ \frac{\partial(\rho u A_p)}{\partial t} + \frac{\partial[(\rho u^2 + p) A_p]}{\partial x} - p \frac{\partial A_p}{\partial x} = \frac{\dot{m}_{ig} A_p \bar{v}_{inj}}{V} + \frac{1}{2} \rho u^2 c_f \\ \frac{\partial(\rho e_0 A_p)}{\partial t} + \frac{\partial(\rho h_0 u A_p)}{\partial x} = \frac{\dot{m}_{pr} A_p H_{pr}}{V} + \frac{\dot{m}_{ig} A_p H_{ig}}{V} + \frac{\dot{m}_{cav} A_p H_{cav}}{V} \end{array} \right.$$

$$\rho = \sum_{k=1}^N \rho_k$$

Base model

Source terms



SPIT Full Model: Cavity Sub-Model

Effective cavity pressure: $\hat{p}_{cav} = p_{cav} - \alpha_{cav}(p_{cav} - p_{cham})$

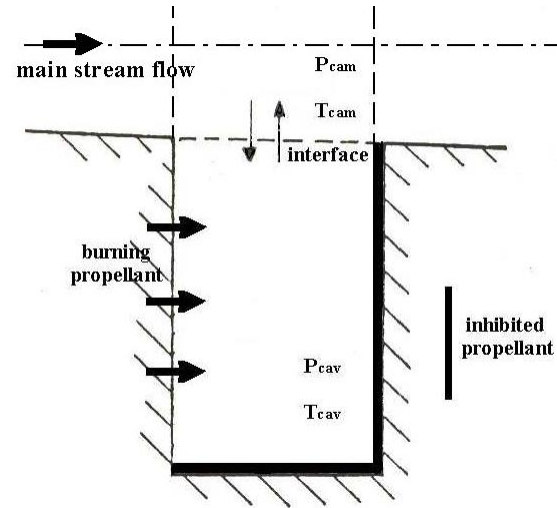
Energy balance within the cavity

$$\Delta H = -(H_{cav} + H_{pr} - \hat{H}_{cav})$$

$$\Delta H < 0 \rightarrow \Delta m_{cav} = \frac{\Delta H}{c_{p,cav} T_{cav}}$$

$$\Delta H > 0 \rightarrow \Delta m_{cav} = \frac{\Delta H}{c_{p,cham} T_{cham}}$$

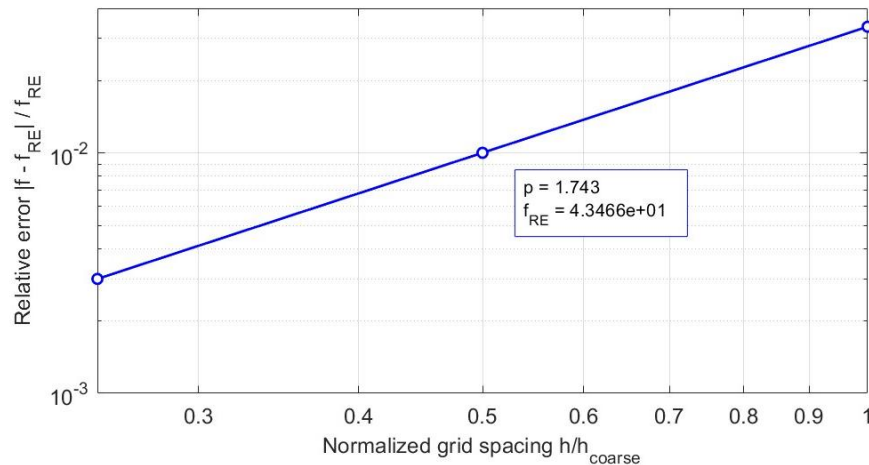
$$\left\{ \begin{array}{l} H_{cav} = c_{p,cav} m_{cav} T_{cav} \\ H_{pr} = r P_b \rho_p dx dt T \\ \hat{H}_{cav} = c_{p,cav} \hat{p}_{cav} \left(\frac{V_{cav}}{R_{cav}} \right) \end{array} \right.$$



CFD Analysis: Domain Discretization and Grid Convergence

Type of igniter	$N_x \times N_y$ of the Igniter(s)	N_{tot}
Axial	80 x 15	183750
Canted	60 x 16	216320
Combined	195 x 20 + 44 x 8	264612

Igniter Nozzle Sub-domain



Global Computational Domain

